Speculative Decoding 690AB

YAML Configurations & Final Results

Overview

This document contains all 9 YAML configurations used for full benchmark runs across the throughput_1k and qualitative datasets, along with the run command and final aggregate metrics for each.

Hardware: 1 × NVIDIA A100-SXM4-40GB on Colab Pro

Target model: meta-llama/Llama-3.1-8B-Instruct

Draft model: meta-llama/Llama-3.2-1B-Instruct

Common settings: gamma=4, max_new_tokens=128, dtype=bfloat16, warmup_steps=5

Quick Reference: Final Results

Throughput_1k (1,536 prompts)

Method	tok/s mean	tok/s median	acceptance	vs baseline
Baseline	26.59	—	—	1.00×
Speculative Greedy	31.15	31.08	0.687	1.17×
AdaSD (tuned)	27.43	26.77	0.640	1.03×
Speculative Stochastic	24.82	24.48	0.523	0.93×
AdaEDL	21.04	20.93	0.400	0.79×

Qualitative (~1,036 rows after multiturn expansion)

Method	tok/s mean	acceptance
Speculative Greedy	30.54	0.663
AdaSD	28.12	0.623
Speculative Stochastic	26.34	0.525
AdaEDL	22.20	0.418

Run Configurations

1. Baseline — Llama 3.1-8B (throughput_1k)

YAML path: `config/baseline_llama8b_1k.yaml`

Configuration


```
method: baseline
target_model: meta-llama/Llama-3.1-8B-Instruct
max_new_tokens: 128
device: cuda
dtype: bfloat16
output: "baseline_llama8b_1k.jsonl"
data: "throughput_1k/0.0.0/41e0d489c7074b1cd4ced3f1ee0e127511c3e091/speed-bench-test.arrow"
warmup_steps: 5
```


Run Command

```
python scripts/benchmark.py --config config/baseline_llama8b_1k.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	26.59	—	n=1536
TTFT (ms)	105	—	
acceptance	—	—	no draft
vs baseline	1.00×	—	reference

2. Speculative Greedy — Llama 1B/8B (throughput_1k)

YAML path: `config/speculative_greedy_llama8b_1k.yaml`

Configuration


```yaml
method: speculative
sampling: greedy
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
output: "speculative_greedy_llama8b_1k.jsonl"
data: "throughput_1k/0.0.0/41e0d489c7074b1cd4ced3f1ee0e127511c3e091/speed-bench-test.arrow"
warmup_steps: 5
```


Run Command

```
python scripts/benchmark.py --config config/speculative_greedy_llama8b_1k.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	31.15	31.08	best speculative method
acceptance	0.687	—	
vs baseline	1.17×	—	

3. Speculative Stochastic — Llama 1B/8B (throughput_1k)

YAML path: `config/speculative_stochastic_llama8b_1k.yaml`

Configuration


```
method: speculative
sampling: speculative
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
temperature: 0.7
output: "speculative_stochastic_llama8b_1k.jsonl"
data: "throughput_1k/0.0.0/41e0d489c7074b1cd4ced3f1ee0e127511c3e091/speed-bench-test.arrow"
warmup_steps: 5
```


Run Command

```
python scripts/benchmark.py --config config/speculative_stochastic_llama8b_1k.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	24.82	24.48	
acceptance	0.523	—	
vs baseline	0.93×	—	underperforms baseline

4. AdaSD (tuned) — Llama 1B/8B (throughput_1k)

YAML path: `config/speculative_ada_llama8b_1k.yaml`

Configuration

```
method: speculative
sampling: ada
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
output: "speculative_ada_llama8b_1k.jsonl"
data: "throughput_1k/0.0.0/41e0d489c7074b1cd4ced3f1ee0e127511c3e091/speed-bench-test.arrow"
warmup_steps: 5
adaptive:
  strategy: ada
  gamma_min: 1
  gamma_max: 8
  step_size: 1
  decrease_factor: 0.5
  smoothing_factor: 0.5
  high_entropy_threshold: 5.0
  avg_da: 0.55
  avg_dr: 0.65
```

Run Command

```
python scripts/benchmark.py --config config/speculative_ada_llama8b_1k.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	27.43	26.77	stdev 6.73, n=1535
acceptance	0.640	0.615	
vs baseline	1.03×	—	tuned thresholds

5. AdaEDL — Llama 1B/8B (throughput_1k)

YAML path: `config/speculative_adaedl_llama8b_1k.yaml`

Configuration

```
method: speculative
sampling: speculative
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
temperature: 0.7
output: "speculative_adaedl_llama8b_1k.jsonl"
data: "throughput_1k/0.0.0/41e0d489c7074b1cd4ced3f1ee0e127511c3e091/speed-bench-test.arrow"
warmup_steps: 5
adaptive:
  strategy: entropy
  gamma_min: 1
  gamma_max: 8
  step_size: 1
  decrease_factor: 0.5
  smoothing_factor: 0.9
  high_entropy_threshold: 5.0
  warmup_steps: 10
  resize: False
```

Run Command

```
python scripts/benchmark.py --config config/speculative_adaedl_llama8b_1k.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	21.04	20.93	
acceptance	0.400	—	
vs baseline	0.79×	—	weakest method

6. Speculative Greedy — Llama 1B/8B (qualitative)

YAML path: `config/speculative_greedy_llama8b_qual.yaml`

Configuration


```
method: speculative
sampling: greedy
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
output: "speculative_greedy_llama8b_qual.jsonl"
data: "qualitative/0.0.0/487aa718444e816458d1a0a52bfce7a454285cf4/speed-bench-test.arrow"
warmup_steps: 5
```


Run Command

```
python scripts/benchmark.py --config config/speculative_greedy_llama8b_qual.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	30.54	—	n=1036 (multiturn)
acceptance	0.663	—	
best category	roleplay	—	32.54 tok/s
worst category	writing	—	26.47 tok/s

7. Speculative Stochastic — Llama 1B/8B (qualitative)

YAML path: `config/speculative_stochastic_llama8b_qual.yaml`

Configuration

```
method: speculative
sampling: speculative
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
temperature: 0.7
output: "speculative_stochastic_llama8b_qual.jsonl"
data: "qualitative/0.0.0/487aa718444e816458d1a0a52bfce7a454285cf4/speed-bench-test.arrow"
warmup_steps: 5
```

Run Command

```
python scripts/benchmark.py --config config/speculative_stochastic_llama8b_qual.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	26.34	—	n=1036
acceptance	0.525	—	
best category	roleplay	—	28.22 tok/s
worst category	writing	—	22.85 tok/s

8. AdaSD — Llama 1B/8B (qualitative)

YAML path: `config/speculative_ada_llama8b_qual.yaml`

Configuration

```
method: speculative
sampling: ada
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
output: "speculative_ada_llama8b_qual.jsonl"
data: "qualitative/0.0.0/487aa718444e816458d1a0a52bfce7a454285cf4/speed-bench-test.arrow"
warmup_steps: 5
adaptive:
  strategy: ada
  gamma_min: 1
  gamma_max: 8
  step_size: 1
  decrease_factor: 0.5
  smoothing_factor: 0.5
  high_entropy_threshold: 5.0
  avg_da: 0.55
  avg_dr: 0.65
```

Run Command

```
python scripts/benchmark.py --config config/speculative_ada_llama8b_qual.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	28.12	—	n=1036
acceptance	0.623	—	
AdaSD wins on	summarization	—	31.84 vs greedy 30.46
struggles on	coding, multilingual	—	23.65–23.92 tok/s

9. AdaEDL — Llama 1B/8B (qualitative)

YAML path: `config/speculative_adaedl_llama8b_qual.yaml`

Configuration


```yaml
method: speculative
sampling: speculative
target_model: meta-llama/Llama-3.1-8B-Instruct
draft_model: meta-llama/Llama-3.2-1B-Instruct
gamma: 4
max_new_tokens: 128
device: cuda
dtype: bfloat16
temperature: 0.7
output: "speculative_adaedl_llama8b_qual.jsonl"
data: "qualitative/0.0.0/487aa718444e816458d1a0a52bfce7a454285cf4/speed-bench-test.arrow"
warmup_steps: 5
adaptive:
  strategy: entropy
  gamma_min: 1
  gamma_max: 8
  step_size: 1
  decrease_factor: 0.5
  smoothing_factor: 0.9
  high_entropy_threshold: 5.0
  warmup_steps: 10
  resize: False
```


Run Command

```
python scripts/benchmark.py --config config/speculative_adaedl_llama8b_qual.yaml
```

Results

Metric	Mean	Median	Notes
tok/s	22.20	—	n=1036, highest variance
acceptance	0.418	—	
best category	coding	—	26.32 tok/s
catastrophic on	humanities	—	16.94 tok/s, acc 0.257